

Ansh Ajaybhai Shah

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EDUCATION

Pandit Deendayal Energy University, Gandhinagar, GUJ

Jun 2024 – Jun 2026

Master of Technology, Data Science CGPA: 8.64 / 10.0

Gandhinagar Institute of Technology, Kalol, GUJ

Jun 2020 – May 2024

Bachelor of Engineering, Information Technology CPI: 8.83 / 10.0

TECHNICAL SKILLS

Languages: Python, C, JavaScript (Node.js)

Machine Learning & Deep Learning: Scikit-Learn, PyTorch, TensorFlow, Neural Networks, MLflow, Optuna, Ray Tune

LLMs & Generative AI: LangChain, Hugging Face Transformers, OpenAI API, Google GenAI SDK, Gemini API, RAG Architecture, Prompt Engineering

Vector Databases: Pinecone, FAISS, Qdrant, Typesense

Big Data & Distributed Systems: Apache Spark, Hadoop, Kafka, Snowflake, Cassandra, Ray

Data Engineering & Pipelines: Apache Airflow, Jenkins, MySQL, MongoDB

Cloud & DevOps: AWS, Azure, Docker, Kubernetes, Prometheus, Grafana

APIs & Web Development: Flask, FastAPI, RESTful APIs, React, Next.js, Tailwind CSS

Automation & Orchestration: n8n

Dashboarding & Visualization: Power BI, Tableau, Qlik Sense, Streamlit, Matplotlib, Seaborn, Plotly, Microsoft Excel

WORK EXPERIENCE

AI Development Intern

Jun 2025 – Jan 2026

Schbang | 7 Months

Multi-Modal Creative Intelligence & RAG-Powered Analytics:

- Architected end-to-end AI pipeline combining Gemini Pro Vision, NLP, and temporal analysis to extract 50+ dimensional features from 500+ ads, achieving 95%+ structured output accuracy.
- Engineered RAG system integrating multi-source consumer data (Reddit, Quora, Google Reviews, Instagram) with brand guidelines and historical performance metrics to generate automated brand audits and data-driven creative recommendations, reducing market research time from weeks to hours.
- Built production-grade AI agents via LangChain with 3-tier retry logic, Pydantic validation, and adaptive error-handling across brand strategy workflows from consumer intelligence extraction to compliance-validated creative generation.

Predictive Compliance & Intelligent Scoring:

- Developed Meta compliance prediction engine achieving 98% accuracy in ad approval prediction, preventing rejected submissions and protecting significant client ad spend.
- Designed dual-category brand scoring system (mass market vs. luxury) with tiered verdicts (Ready to Push / Needs Changes / Rethought) and optimization recommendations that directly improved client ROAS.

Behavioral Analytics & Consumer Segmentation:

- Engineered consumer profiling framework with 35+ behavioral dimensions and 300+ enumerated patterns across D2C and B2B contexts, applying clustering and statistical modeling to identify distinct consumer cohorts.
- Built ad-to-audience alignment engine using ML to predict optimal creative-segment pairings, reducing creative testing cycles by 40% through data-driven segment analysis and visualization dashboards.

AI-Generated Creative Production:

- Developed adaptive creative iteration system transforming underperforming ads into high-performing assets with Meta Andromeda compliance for both image and video formats.
- Implemented concept-to-creative pipeline using Gemini image generation with integrated compliance validation and quality scoring; pipeline assets were deployed in live campaigns achieving measurable gains in reach and ROAS.
- Contributed to frontend of getadvize.ai using React, Next.js, and Tailwind CSS, building responsive orientation-aware UI components.

Tech Stack: Gemini (Pro Vision, 2.5 Pro/Flash), LangChain, Python, FastAPI, Pydantic, RAG, Pinecone, Computer Vision, NLP, Streamlit, Plotly, Next.js, Tailwind CSS

Freelance Data Scientist

Oct 2024 – Feb 2025

Aashirvad Chemicals | Remote | 5 Months

- Built predictive models to optimize chemical formulations and improve batch yield using Python (pandas, scikit-learn).

- Analyzed production data to identify key variables affecting quality, reducing material waste and process variability.
- Automated anomaly detection and reporting pipelines with Twilio-based real-time SMS alert integration.
- Developed Streamlit dashboards to visualize KPIs and support data-driven decision-making for plant operations.

Data Science Intern

Apr 2024 – May 2024

BrainyBeam Technologies Pvt Ltd | 2 Months

- Built and evaluated supervised ML models (Logistic Regression, Random Forest) using Python and Scikit-Learn.
- Performed EDA, data cleaning, and feature engineering; improved model performance via cross-validation and hyperparameter tuning.
- Implemented NLP workflows (text preprocessing, TF-IDF, classification) for small-scale text datasets.
- Presented model results and insights through visualizations and summary reports.

PROJECT EXPERIENCE

1. News Sentiment Analyzer

Stack: Python (FastAPI), Streamlit, BERT, NewsAPI, BeautifulSoup, Pydantic, gTTS, Docker

- Architected full-stack microservices application with FastAPI backend and Streamlit frontend delivering real-time financial sentiment forecasting.
- Engineered automated data pipeline scraping live news via NewsAPI and BeautifulSoup with Pandas-based preprocessing; deployed BERT transformer for high-precision sentiment classification (Positive / Negative / Neutral) on financial headlines.
- Built multi-modal output layer featuring AI-driven TTS (gTTS) and async Uvicorn endpoints with Pydantic validation; containerized with Docker for portable, reproducible deployment across cloud environments.

2. Thesis RAG System: GraphRAG-Enhanced Research Chatbot for Ophthalmic AI Literature

Stack: Python, FastAPI, LangChain LCEL, Qdrant (HNSW), BGE-large-en-v1.5, rank_bm25, Redis, NetworkX, Grobid, PyMuPDF, Camelot, spaCy, LLMingua-2, Claude 3.5 Sonnet, Docker Compose, Nginx, Streamlit

- Built production-grade GraphRAG-enhanced hybrid retrieval pipeline over a 6-chapter thesis and 30 research papers using BGE-large-en-v1.5 (1024-dim) embeddings merged with BM25 via Reciprocal Rank Fusion (alpha=0.6 vector, 0.4 BM25), enabling entity-centric queries 40% faster than Agentic RAG by replacing multi-step tool-call loops with direct in-memory graph traversal.
- Engineered 5-stage query pipeline (LLM-based routing, HyDE hypothetical-document expansion, parallel hybrid retrieval via LangChain LCEL RunnableParallel, BGE cross-encoder reranking top-20 to top-5, and LLMingua-2 context compression at ratio=0.5) achieving RAGAS faithfulness 0.85+, context precision 0.80+, and context recall 0.80+ across equation_lookup, comparative, synthesis, and factual query types.
- Designed automated ingestion pipeline parsing research PDFs with Grobid (structured section and equation extraction) and PyMuPDF, followed by MinHash LSH deduplication (threshold=0.85, 128 permutations via datasketch) and LangChain SemanticChunker with parent-child hierarchy, delivering clean non-redundant chunks enriched with 12-field Pydantic metadata schema for pre-retrieval filtering.
- Constructed NetworkX knowledge graph with 8 relation types (uses_algorithm, evaluates_on_disease, trained_on_dataset, achieves_metric) using spaCy PhraseMatcher across a 5-domain entity taxonomy covering bio-inspired algorithms, DL architectures, diseases, datasets, and metrics, enabling sub-millisecond entity queries without vector search overhead.
- Implemented Redis dual-cache strategy (query cache TTL=24h, embedding cache TTL=7d keyed by SHA-256 hash) with per-session ConversationBufferWindowMemory (k=5 turns) backed by RedisChatMessageHistory, eliminating redundant LLM API calls while maintaining coherent multi-turn conversational context.
- Deployed containerized system via Docker Compose (FastAPI + Qdrant + Redis + Nginx) with structured JSON logging via structlog, per-user API key authentication, rate limiting middleware, and GitHub Actions CI/CD pipeline, ensuring reproducible production-ready deployment across environments.